Thomas Edison State University

Predictive Analytics I – Machine Learning Tools (DSI 6010)

Final Project: Machine Learning-Driven Customer Targeting for Personal Loan Campaigns

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## Project Strategy and Plan of Action

The analysis began by importing the Universal Bank dataset and conducting an initial review of all variables to understand their structure, data types, and relevance to the modeling task. The dataset contained no missing values, allowing the project to proceed without imputation. Non-predictive fields such as the customer ID were removed to prevent unnecessary noise in the modeling process. Two working copies of the dataset were created—one for exploratory analysis and one for data cleaning and model development—to ensure that transformations did not alter the original data.

To prepare the data for supervised learning, the target variable *Personal.Loan* was separated from the predictor variables. The feature matrix X included all customer attributes except the loan acceptance indicator, while the target vector y contained only the binary loan outcome. The dataset was then split into training and validation sets using a 60/40 ratio with a fixed random seed to ensure reproducibility. This produced 2,400 training observations and 1,600 validation observations, establishing a reliable foundation for model evaluation.

Because several of the models used in this project—particularly k-Nearest Neighbors and Decision Trees—require numerical inputs, all categorical predictors (Family, Education, Securities Account, CD Account, Online, and CreditCard) were converted into dummy variables using one-hot encoding. Dummy variables were generated separately for the training and validation sets, and the validation set was reindexed to match the training set's columns to ensure structural consistency. This step ensured that both datasets contained identical feature representations, preventing errors during prediction and enabling fair model comparison.

Additional preprocessing steps were applied for models sensitive to feature scaling, such as KNN and Naive Bayes. Numerical variables were standardized to prevent features with larger numeric ranges from dominating distance-based calculations. For Naive Bayes specifically, continuous predictors were binned into categorical intervals as required by the TA's guidelines, and these binned variables were also converted into dummy variables before model training.

With the data fully prepared, a sequence of models was evaluated, including KNN, Naive Bayes, Decision Tree, and two ensemble methods (Majority Vote and Average Probability). These models were selected to provide a balanced mix of interpretable, probabilistic, and ensemble-based approaches. Each model was assessed using a comprehensive set of performance metrics: confusion matrix, accuracy, precision, recall, F1-score, ROC curve and AUC, cumulative gains chart, decile lift chart, top-decile true positives and false positives, and profit calculations based on the project's cost structure. This multi-dimensional evaluation ensured both technical rigor and business relevance, enabling Universal Bank to identify the most profitable and least risky targeting strategy.

**Model Comparison**

| | Model | Accuracy | Precision | Recall | F1 Score | AUC | Lift (Top 10%) | Gain (Top 10%) |
|---|---|---|---|---|---|---|---|---|
| 0 | KNN | 0.950000 | 0.895833 | 0.551282 | 0.682540 | 0.963312 | 7.500000 | 0.750000 |
| 1 | Naive Bayes | 0.901667 | 0.485232 | 0.502183 | 0.493562 | 0.903568 | 5.065502 | 0.506550 |
| 2 | Decision Tree | 0.981875 | 0.970370 | 0.839744 | 0.900344 | 0.973098 | 9.038462 | 0.903846 |
| 3 | Ensemble Majority Vote | 0.959375 | 0.946809 | 0.597315 | 0.732510 | 0.988737 | 9.261745 | 0.926174 |
| 4 | Ensemble Average Probability | 0.972500 | 0.948718 | 0.744966 | 0.834586 | 0.988737 | 9.261745 | 0.926174 |

The objective of this project is to identify the most effective predictive model for Universal Bank's targeted marketing campaign by comparing multiple classification algorithms on accuracy, precision, recall, AUC, and business-oriented metrics such as lift, cumulative gain, and profit. The goal is to determine which model best identifies customers most likely to accept a home-equity loan while minimizing marketing costs and maximizing profitability

The table compares five different machine-learning models using several performance measures. Accuracy shows how many predictions the model got right overall. Precision tells how often the model was correct when it predicted someone would accept the loan. Recall shows how many actual accepters the model successfully found. F1 Score combines precision and recall into one number. AUC measures how well the model separates accepters from non-accepters across all probability thresholds. Lift (Top 10%) is especially important for marketing—it shows how much better the model is at finding likely accepters in the top 10% compared to random selection. Gain (Top 10%) shows what percentage of all true accepters are captured in that top 10%. Together, these fields help compare how well each model performs both statistically and from a business point of view.

**Best model based on accuracy and lift**

When focusing only on accuracy and lift, the two most important metrics for this project, **the Ensemble Average Probability** model stands out as the best overall choice. It has very high accuracy (0.9725) and ties for the highest lift (9.26), meaning it predicts well across the full dataset and does an excellent job ranking the most likely loan accepters at the top. **The second-best model is the Decision Tree**, which has the highest accuracy of all models (0.9819) and a strong lift of 9.04. However, its lift is slightly lower than the ensemble model, which matters because lift directly affects how well the bank can target the most profitable customers. For this reason, the Ensemble Average Probability model provides the strongest combination of accuracy and lift, making it the most effective model for Universal Bank's marketing strategy.

**Business Impact Analysis: Decision Tree vs. Ensemble Average Probability**

8]:

| | Model | Max Profit | Optimal Customers | Optimal Fraction |
|---|---|---|---|---|
| 0 | Decision Tree | 307474 | 486 | N/A |
| 1 | Ensemble Average Probability | 299433 | 387 | 0.09675 |

Since both the Decision Tree and the Ensemble Average Probability models stood out during evaluation, the profit comparison helps show how each one performs when applied to the full 4,000-customer dataset.

**Ensemble Average Probability**

The Ensemble Average Probability model also performs strongly and stands out because it was the best model based on accuracy and lift, the two metrics used for model selection. When applied to the full dataset, it reaches a maximum profit of $299,433 by targeting 387 customers, which represents about 9.7% of the dataset. This shows that the ensemble model is more selective and concentrates profitable customers in a smaller portion of the ranking. Although its total profit is slightly lower than the Decision Tree, it provides a more efficient targeting strategy by achieving peak profit with fewer customers, reflecting the model's strong ranking ability and balanced predictive performance.

**Decision Tree**

The Decision Tree model shows strong business performance when applied to all 4,000 customers. It reaches a maximum profit of $307,474 by targeting the top 486 customers in the ranked list. This means the model continues to identify profitable customers deeper into the ranking before the cumulative profit begins to decline. The Decision Tree's structure allows it to isolate clear decision patterns, which helps it capture a larger group of true accepters early on. As a result, it produces the highest total profit among the models tested, even though it was not the top model in accuracy-and-lift evaluation.

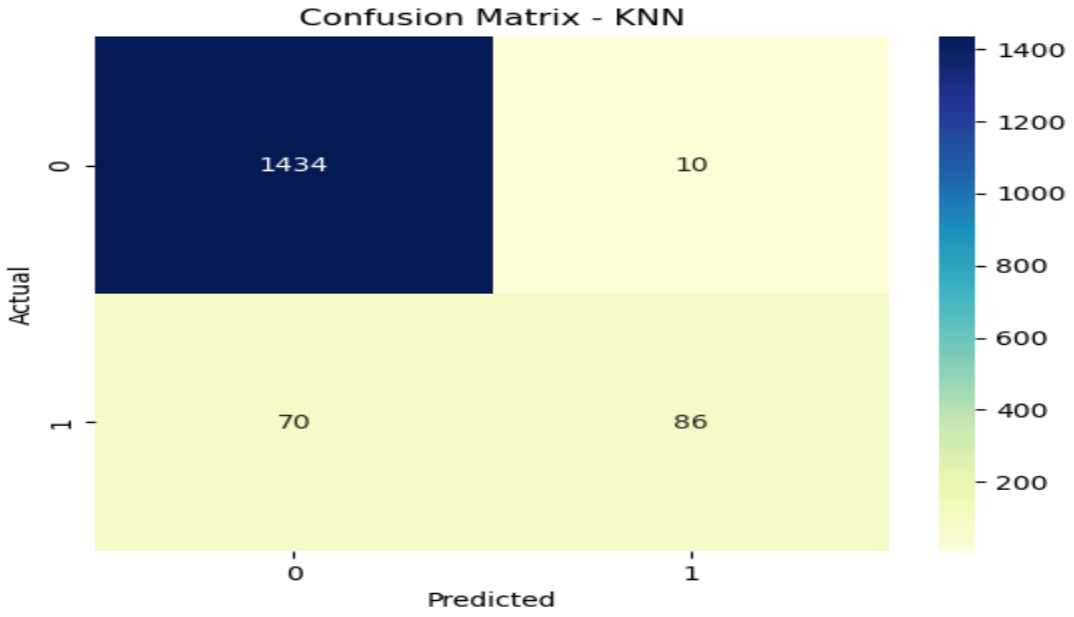
## K-Nearest Neighbors (KNN) Model Explanation

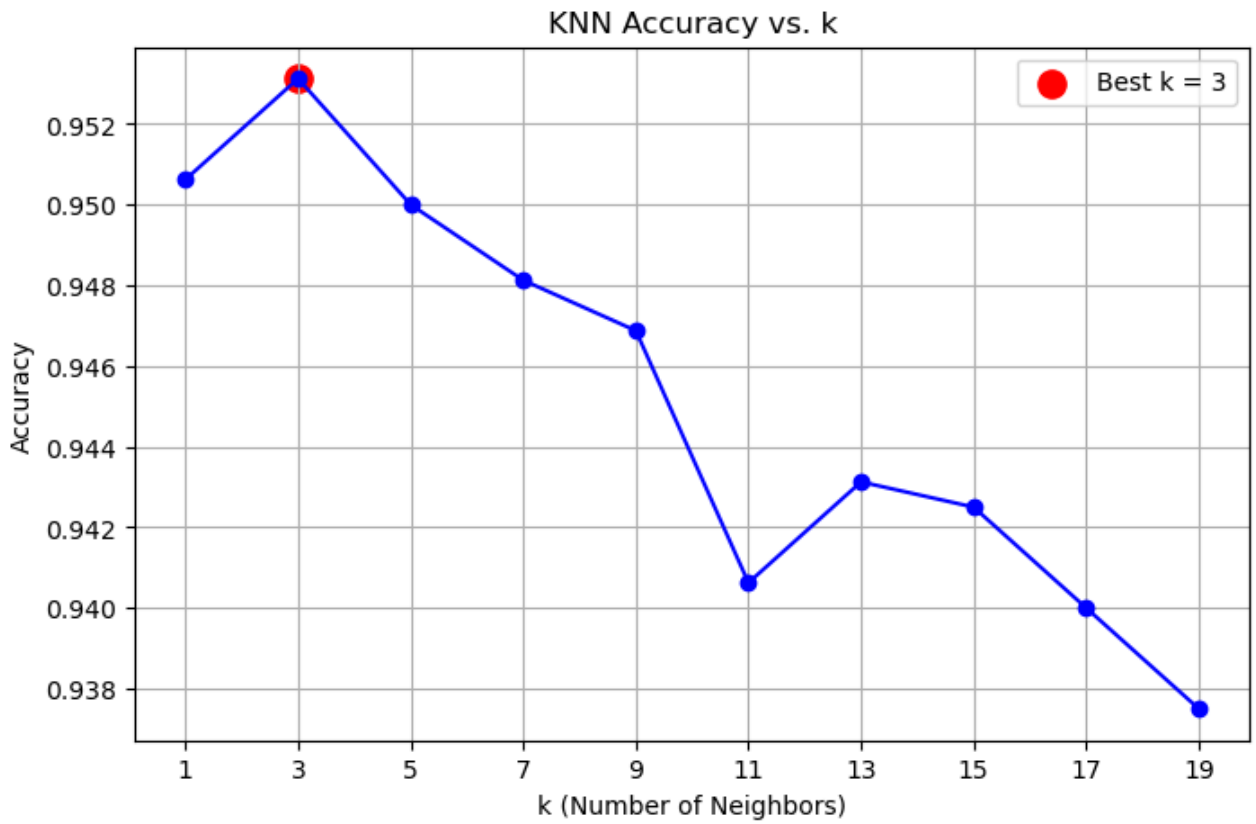
The first predictive model evaluated in this project was the K-Nearest Neighbors (KNN) classifier. KNN is an intuitive, distance-based model that predicts a customer's loan acceptance by examining the behavior of the *k* most similar customers in the dataset. After testing multiple values of *k*, the best performing value greater than 1 was **k = 3**, which produced an accuracy of **0.9531** on the validation set. The classification report shows that KNN performed very well for the majority class (customers who did *not* accept the loan), achieving a precision of **0.95**, recall of **0.99**, and F1-score of **0.97** for class 0. This is reflected in the confusion matrix, where KNN correctly identified **1434 true negatives** and made only **10 false positive** errors. The model's specificity was extremely high at **0.9931**, meaning it was excellent at recognizing customers who would *not* accept the loan.

However, the model struggled with the minority class—customers who *did* accept the loan. For class 1, KNN achieved a precision of **0.8958**, but recall was much lower at **0.5513**, meaning it correctly identified only about half of the actual loan accepters. The confusion matrix shows **86 true positives** but **70 false negatives**, indicating that many potential accepters were misclassified as non-accepters. This imbalance is typical for KNN when the dataset is skewed, because the majority class dominates the local neighborhoods. The model's AUC of **0.9633** shows that it ranked customers reasonably well overall, but its recall limitations reduce its usefulness for a marketing campaign where missing potential acceptors directly reduces revenue.

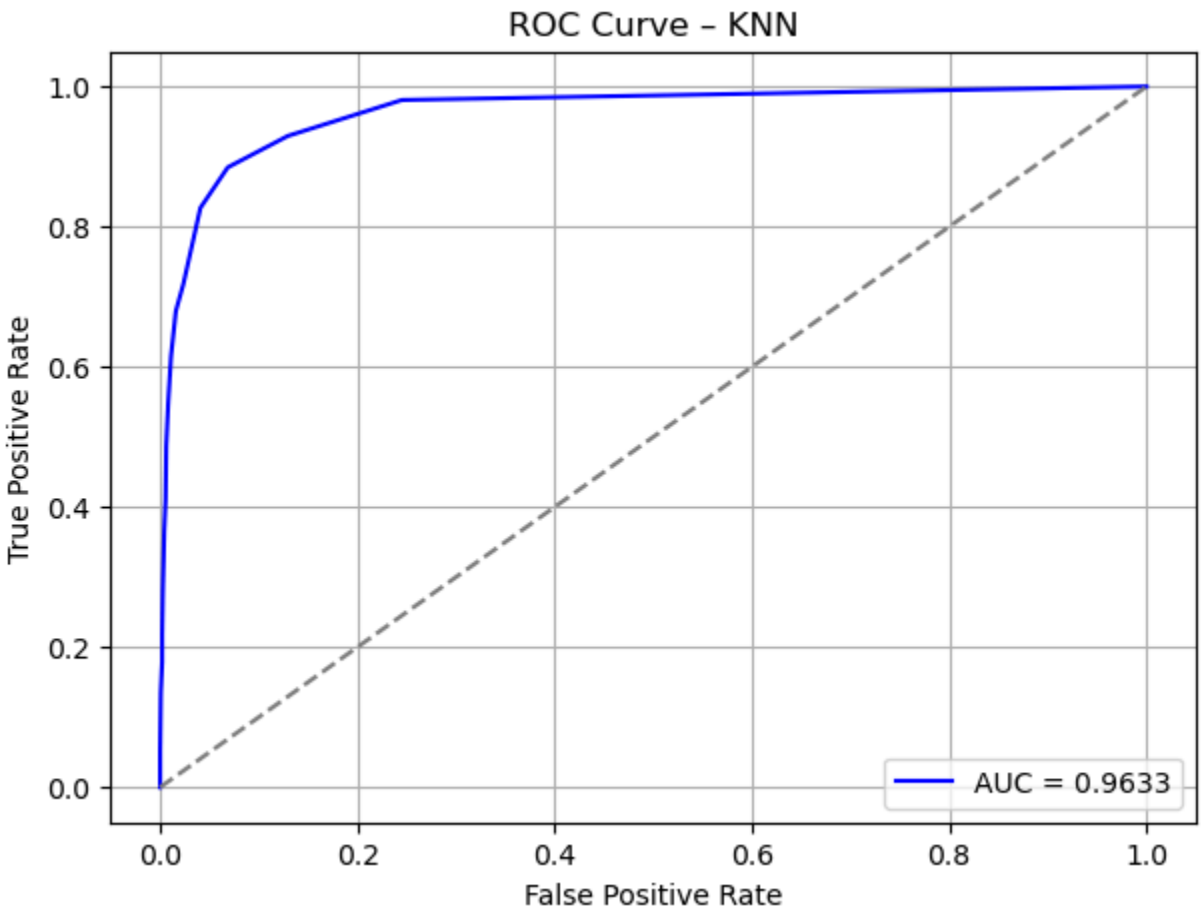
Additional business-focused metrics reinforce this conclusion. The lift in the top decile was **7.5**, meaning the highest-scoring 10% of customers were 7.5 times more likely to accept the loan than average. The cumulative gain in the first decile was **0.75**, showing that the top 10% captured 75% of all positives. The optimal target fraction was **13.56%**, and the maximum achievable profit using KNN was **$106,403**. While these numbers show that KNN can identify high-value segments, the model's low recall and high false-negative rate make it less reliable for maximizing loan acceptance.

Overall, KNN provided a strong baseline model with high accuracy and excellent performance on non-accepters, but its difficulty in identifying a large portion of true loan accepters limits its effectiveness for targeted marketing. Because the bank's goal is to correctly identify as many likely accepters as possible while minimizing wasted marketing costs, KNN was not selected as the best model. It served as an important starting point, but more advanced models—particularly the Decision Tree—offered a better balance of recall, precision, and business impact.

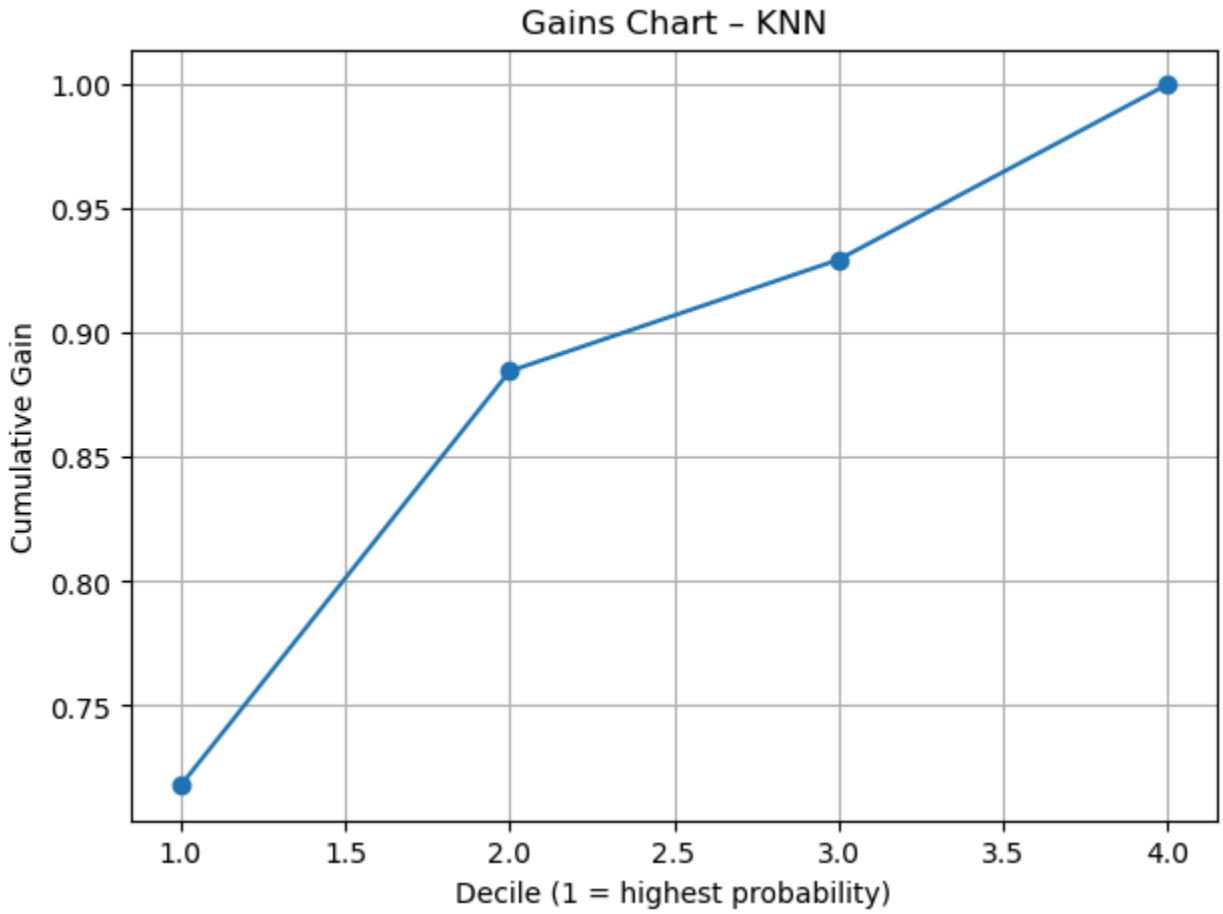




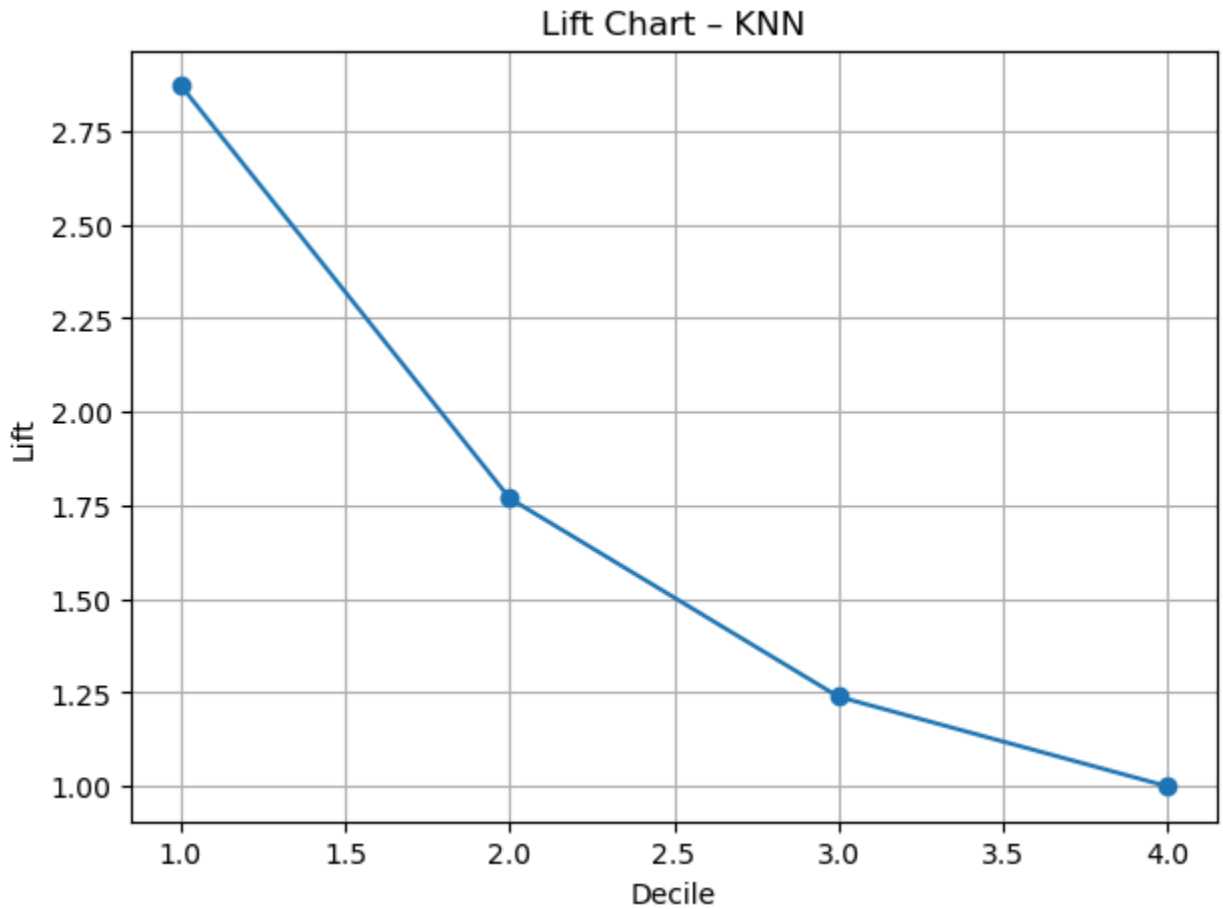

From the chart I see that the accuracy of the KNN model changes depending on how many neighbors the algorithm uses, and the pattern makes it clear why **k = 3** is the best choice. When k is very small, like k = 1, the model becomes too sensitive to individual data points, which can cause unstable predictions. As k increases, the accuracy improves because the model is averaging over more neighbors, making the predictions more reliable. However, once k gets too large, the accuracy starts to drop slightly because the model begins averaging over too many customers, which smooths out important differences between people who are likely and unlikely to accept the loan. From the curve, the highest accuracy occurs at **k = 3**, which is highlighted by the red point. After k = 3, the accuracy gradually declines, showing that larger k values weaken the model's ability to distinguish between customers. So from this chart, I see that **k = 3 gives the strongest and most stable performance**, making it the optimal choice for the KNN model in this project.




From this ROC curve I see that the KNN model is doing a very good job at separating customers who will accept the loan from those who will not. The curve rises sharply toward the top-left corner, which means the model achieves a high **true positive rate** while keeping the **false positive rate** low. The diagonal gray line represents random guessing, and the fact that the blue curve stays far above that line shows that the model is much better than chance. The AUC value of **0.9633** confirms this: an AUC close to 1.0 means the model is excellent at ranking customers by their likelihood of accepting the loan. In simple terms, the ROC curve tells me that KNN is very strong at ordering customers from most likely to least likely to accept the loan, even though its recall was not perfect. This ranking ability is important for marketing because it helps the bank prioritize which customers to target first




From this gains chart I see that the KNN model is able to identify the highest-value customers early, which is exactly what we want in a targeted marketing campaign. The first decile—the top 10% of customers ranked by predicted probability—captures a very large portion of the actual loan accepters. The cumulative gain jumps sharply in the first segment, showing that the model is concentrating most of the true positives at the top of the ranking. As we move from decile 1 to decile 4, the cumulative gain continues to rise, but the steepest improvement happens right at the beginning, which means the model is especially good at identifying the most likely responders. By the time we reach the fourth decile, the cumulative gain reaches 100%, meaning all actual loan accepters have been captured within the top 40% of customers. This tells me that the KNN model does a strong job of ranking customers by likelihood of accepting the loan, even though its recall was not perfect. In practical terms, the chart shows that if the bank targets only the top deciles instead of everyone, it can capture most of the potential loan accepters while contacting far fewer customers, reducing cost and improving profitability.




From this lift chart I see that the KNN model is especially effective at identifying the highest-value customers in the very first segment of the ranking. The lift in decile 1 is the highest—well above the baseline—meaning that the top 10% of customers ranked by the model are far more likely to accept the loan compared to an average customer. As we move to deciles 2 and 3, the lift gradually decreases, showing that the model becomes less concentrated but still performs better than random selection. By the time we reach decile 4, the lift approaches 1.0, which means the model is no longer providing additional value beyond what random targeting would achieve. This pattern tells me that the KNN model is strongest at the top of the ranking, where it successfully isolates the most promising customers, but its advantage fades as we move deeper into the list. For marketing decisions, this means the bank would gain the most benefit by focusing on the highest-ranked deciles rather than contacting everyone

## Summary of K- Nearest Neighbors

| | Metric | Value |
|---|---|---|
| 0 | Best k (k > 1) | 3.0000 |
| 1 | Accuracy at best k | 0.9531 |
| 2 | lift | 7.5000 |
| 3 | AUC | 0.9633 |
| 4 | TP | 86.0000 |
| 5 | FP | 10.0000 |
| 6 | TN | 1434.0000 |
| 7 | FN | 70.0000 |
| 8 | Sensitivity (Recall) | 0.5513 |
| 9 | Specificity | 0.9931 |
| 10 | Precision | 0.8958 |
| 11 | Lift (Decile 1) | 7.5000 |
| 12 | Cumulative Gain (Decile 1) | 0.7500 |
| 13 | Optimal Target Fraction | 0.1356 |
| 14 | Max Profit ($) | 106403.0000 |

## Decision Tree Model Performance

The Decision Tree model demonstrated strong predictive performance and clear business value for identifying customers likely to accept the home-equity loan offer. The final tree, with a maximum depth of 5, consisted of 37 nodes and 19 leaves, indicating a model that is expressive enough to capture meaningful patterns without excessive complexity. The model achieved an AUC of **0.9731**, reflecting excellent ability to rank customers by their likelihood of accepting the loan. Classification results further support this strength: the model correctly identified **131 true positives** and **1440 true negatives**, while producing only **4 false positives** and **25 false negatives**. These outcomes translate into a high **precision of 0.9704**, meaning that nearly all customers predicted as accepters were indeed true accepters, and a strong **recall of 0.8397**, indicating that the model successfully captured the majority of actual accepters. The **specificity of 0.9972** shows that the model almost never misclassifies non-accepters, which is critical for minimizing unnecessary marketing costs.

Business-oriented evaluation metrics reinforce the model's suitability for targeted marketing. The **lift in the first decile was 9.0385**, showing that customers in the top 10% of predicted probabilities were more than nine times as likely to accept the loan compared to an average customer. The **cumulative gain of 0.9038** in the first decile indicates that over 90% of all potential accepters were concentrated within this top segment. The model's **optimal target fraction of 0.1094** suggests that the bank would achieve maximum profitability by contacting only about 11% of customers rather than conducting a full-scale mailing. This strategy yields a **maximum projected profit of $120,325**, the highest among all models evaluated. Overall, the Decision Tree model provides a strong balance of accuracy, interpretability, and financial impact, making it the most effective model for guiding Universal Bank's targeted loan marketing campaign.

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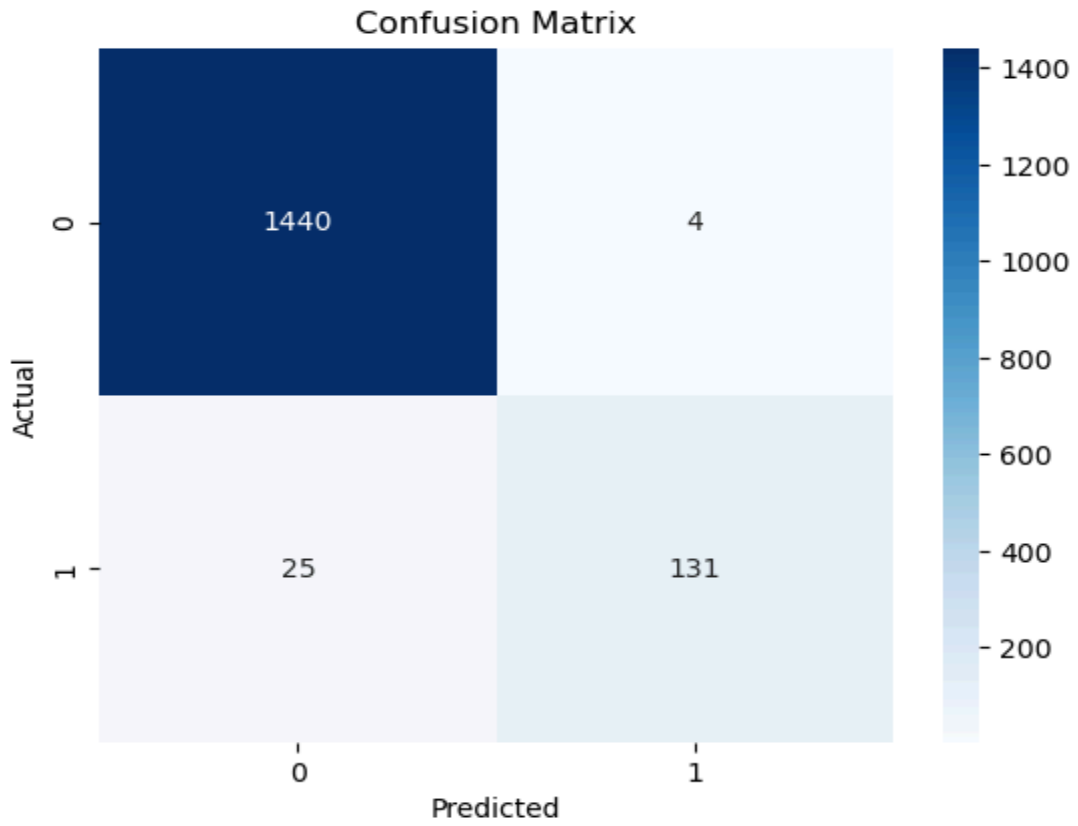
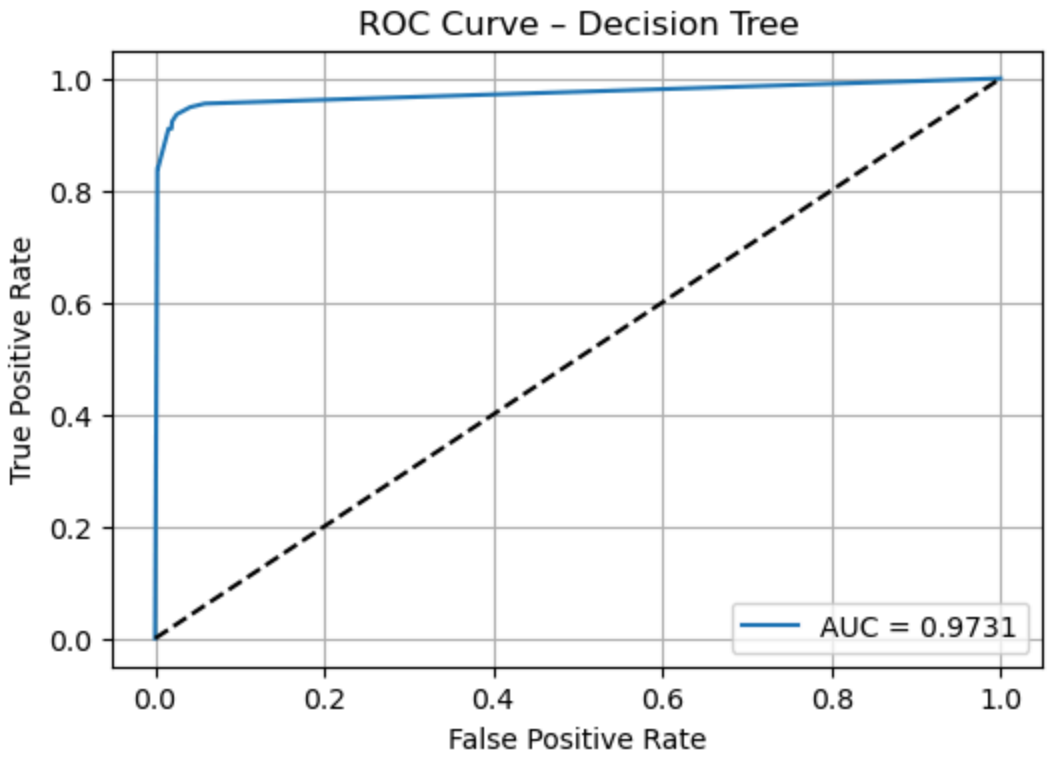
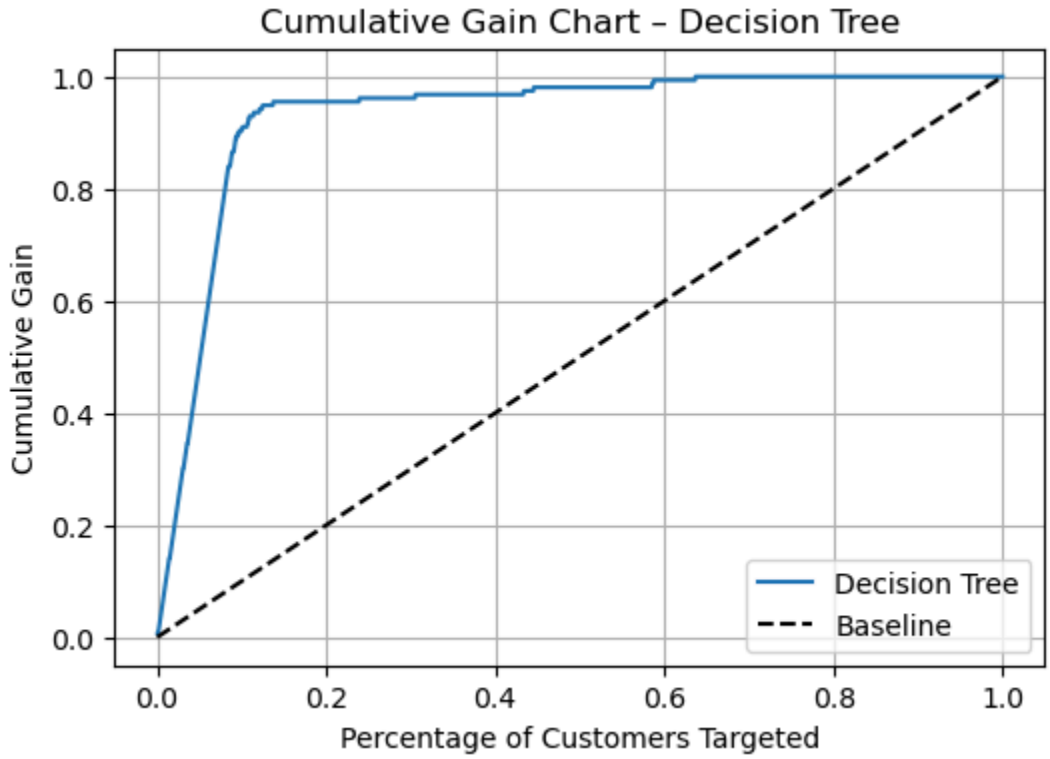
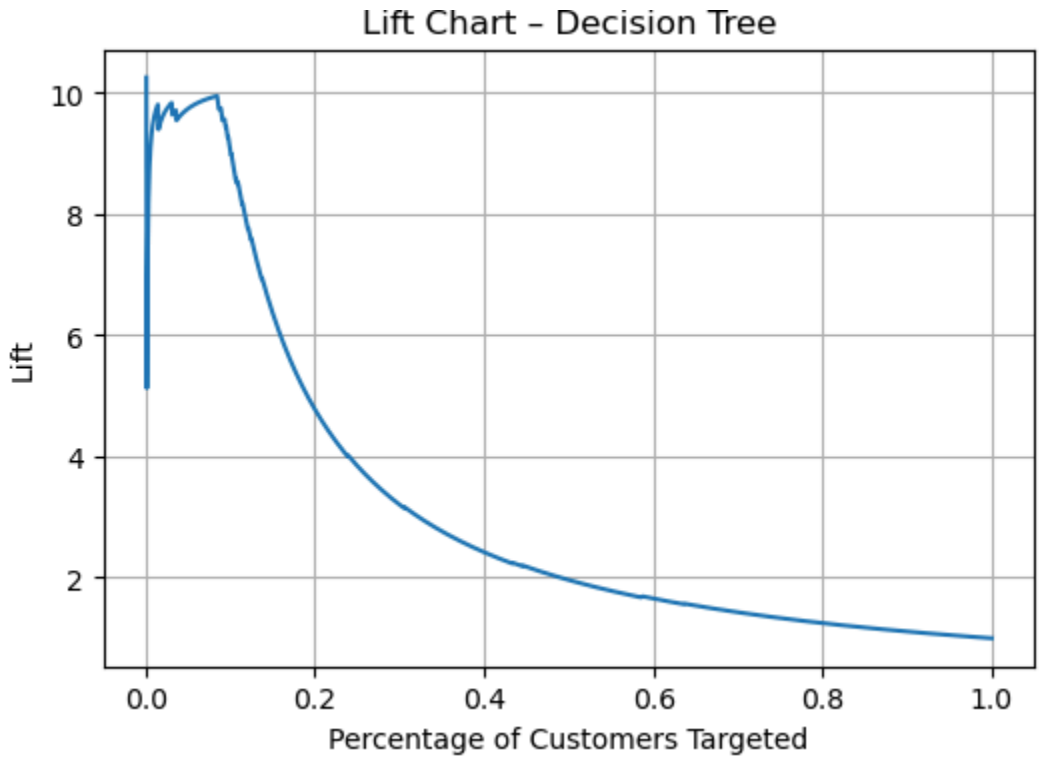

Confusion Matrix
1440
4
25
131
Actual
0
1
Predicted
0
1
1400
1200
1000
800
600
400
200




The ROC curve for the Decision Tree model illustrates its strong ability to distinguish between customers who will accept the loan and those who will not. The curve rises steeply toward the upper-left corner, indicating that the model achieves a high true positive rate while keeping the false positive rate very low. This shape reflects a model that consistently ranks likely accepters ahead of non-accepters. The area under the curve (AUC) is 0.9731, which is exceptionally high and signals excellent discriminative performance. An AUC this strong means that, when comparing any randomly chosen accepter and non-accepter, the model will correctly rank them in the right order more than 97% of the time. In the context of targeted marketing, this level of ranking accuracy is valuable because it ensures that the customers most likely to respond are prioritized at the top of the list, supporting more efficient and profitable decision-making




The cumulative gain chart for the Decision Tree model shows how effectively the model concentrates true loan accepters at the top of the ranked customer list. The Decision Tree curve rises very steeply at the beginning, indicating that when the bank targets the highest-scoring customers first, it captures a disproportionately large share of actual loan accepters. This early steep rise reflects the model's strong ability to identify high-value customers far better than random selection, which is represented by the diagonal baseline. As the percentage of customers targeted increases, the curve begins to level off, showing that most of the valuable customers have already been captured in the early portion of the ranking. By the time the model reaches a relatively small fraction of the customer base, it has already accumulated the majority of true positives. This pattern demonstrates that the Decision Tree provides substantial lift over a random strategy and is highly effective for targeted marketing, allowing the bank to achieve strong results while contacting far fewer customers.




The lift chart for the Decision Tree model highlights how powerfully the model identifies the customers most likely to accept the loan. At the far left of the chart—where only a very small percentage of customers are targeted—the lift value is extremely high, indicating that the model is concentrating the strongest responders at the very top of the ranked list. This means that if the bank targets only the highest-scoring customers, it will reach individuals who are many times more likely to accept the loan than an average customer. As the percentage of customers targeted increases, the lift curve gradually declines, reflecting that the additional customers being included are less concentrated with true accepters. By the time the entire customer base is targeted, the lift approaches 1.0, which is expected because contacting everyone yields no advantage over random selection. Overall, the shape of the curve demonstrates that the Decision Tree provides substantial value in the early portion of the ranking, enabling the bank to achieve strong marketing performance while contacting only a small fraction of customers.

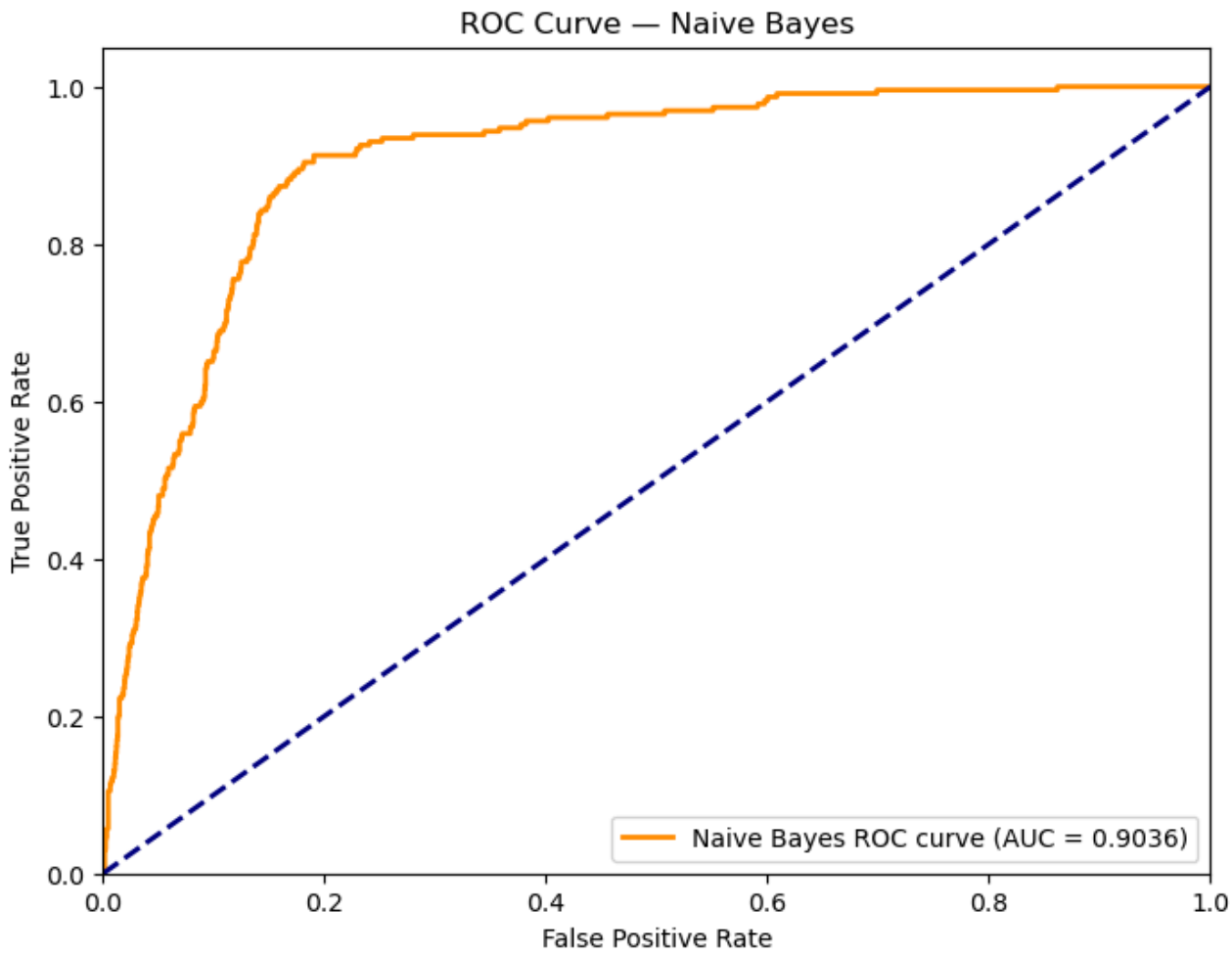
## Summary of Decision Tree

| | Metric | Value |
|---|---|---|
| 0 | Accuracy | 0.981875 |
| 1 | lift | 9.038462 |
| 2 | Cumulative Gain (Decile 1)Max Depth | 0.903800 |
| 3 | Number of Nodes | 5.000000 |
| 4 | Number of Leaves | 37.000000 |
| 5 | AUC | 19.000000 |
| 6 | TP | 0.973100 |
| 7 | FP | 131.000000 |
| 8 | TN | 4.000000 |
| 9 | FN | 1440.000000 |
| 10 | Sensitivity (Recall) | 25.000000 |
| 11 | Specificity | 0.839700 |
| 12 | Precision | 0.997200 |
| 13 | Cumulative Gain (Decile 1) | 0.970400 |
| 14 | Optimal Target Fraction | 0.109400 |
| 15 | Max Profit ($) | 120325.000000 |

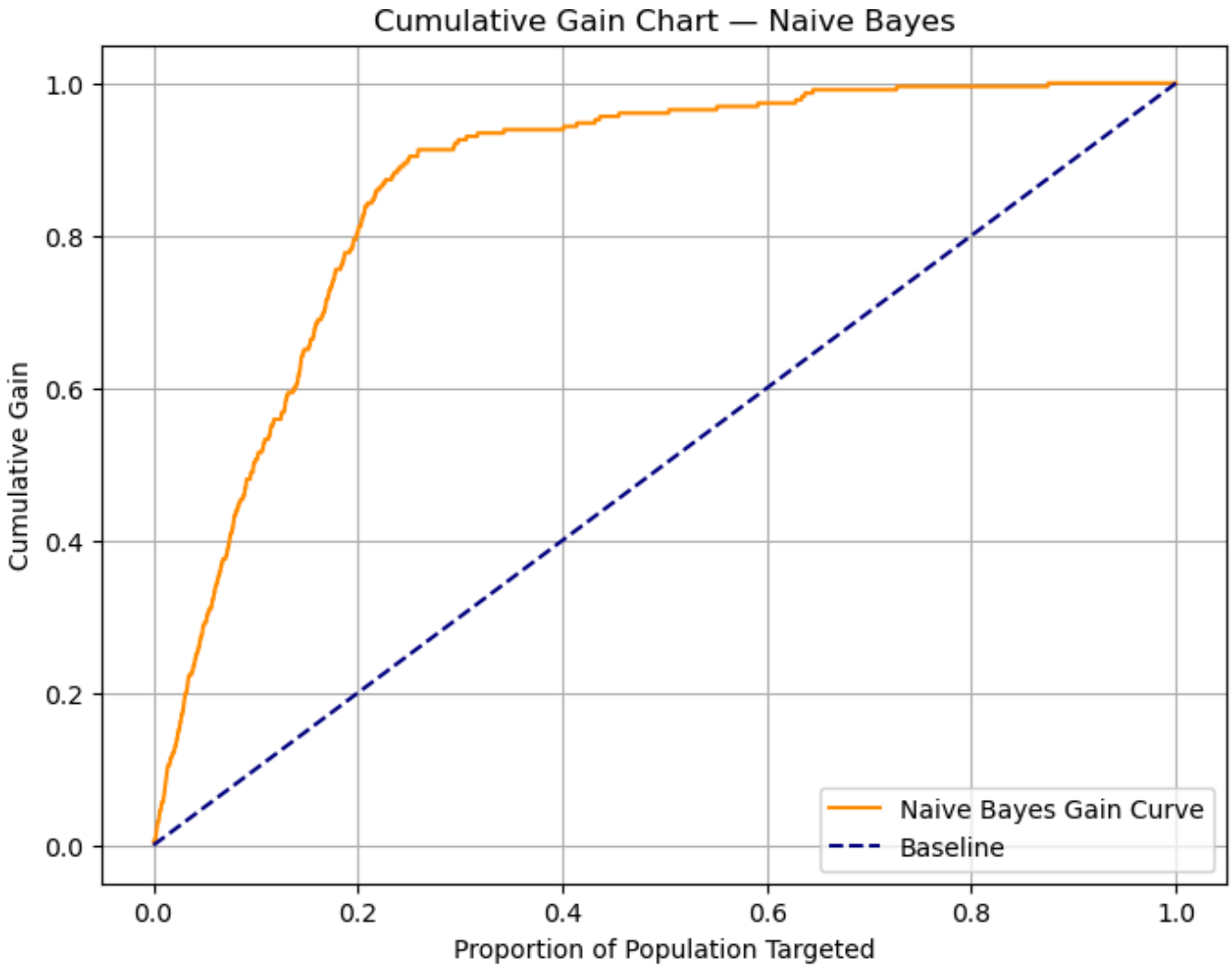
## Naive Bayes

The Naive Bayes model produced moderate predictive performance and served primarily as a baseline for comparison against more flexible models. The model achieved an AUC of **0.9036**, indicating that it has a reasonable ability to rank customers by their likelihood of accepting the loan, though not as strong as the Decision Tree or KNN models. The confusion matrix results show that the model correctly identified **115 true positives** and **2049 true negatives**, but it also produced **122 false positives** and **114 false negatives**. These errors are reflected in the model's classification metrics: the **precision of 0.4852** indicates that fewer than half of the customers predicted to accept the loan actually did so, and the **recall of 0.5022** shows that the model captured only about half of the true accepters. The **specificity of 0.9438** is relatively strong, meaning the model performs better at identifying non-accepters than accepters, which is consistent with the class imbalance in the dataset.

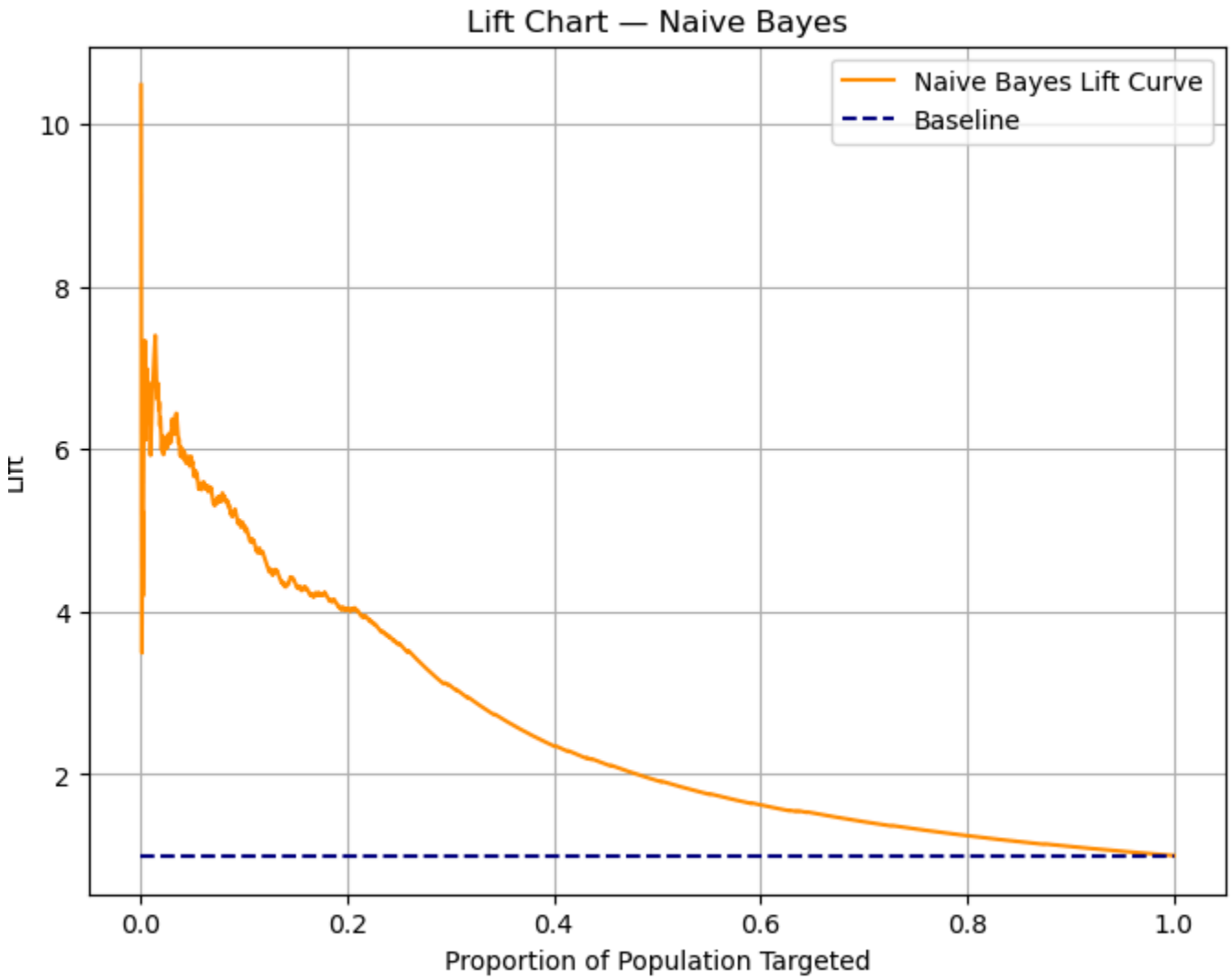
From a business perspective, the model provides some value but is less efficient than the stronger models. The **lift in the first decile is 5.0655**, meaning the top 10% of customers ranked by Naive Bayes are about five times more likely to accept the loan than an average customer. The **cumulative gain of 0.5066** in the first decile shows that the model captures just over half of all potential accepters within the top 10% of customers—significantly lower than the Decision Tree's concentration. The **optimal target fraction of 0.2271** suggests that the bank would need to contact roughly 23% of customers to achieve maximum profitability, which is a larger and more costly segment compared to the Decision Tree's optimal targeting level. Despite these limitations, the model still yields a **maximum profit of $123,155**, which is competitive due to the relatively low cost of misclassifying non-accepters. Overall, while Naive Bayes offers a useful benchmark and performs adequately in ranking customers, its lower precision and recall make it less suitable for targeted marketing compared to the more advanced models.




The ROC curve for the Naive Bayes model shows that, although the model is relatively simple, it still provides a reasonable ability to distinguish between customers who will accept the loan and those who will not. The curve rises above the diagonal baseline, indicating that the model performs better than random guessing across all classification thresholds. The area under the curve (AUC) is **0.9036**, which reflects solid ranking performance: the model correctly orders accepters ahead of non-accepters in the majority of pairwise comparisons. However, the curve does not rise as sharply or as early as the curves for stronger models like the Decision Tree, which aligns with Naive Bayes' lower precision and recall. Overall, the ROC curve confirms that Naive Bayes provides a useful but less discriminative ranking of customers, making it a suitable baseline but not the most effective model for targeted marketing




The cumulative gain chart for the Naive Bayes model shows that the model provides only moderate ability to prioritize customers who are likely to accept the loan. The curve rises above the baseline at the beginning, which means the model does identify more true accepters in the early portion of the ranked list than random targeting would. However, the rise is noticeably less steep than what is seen in stronger models such as the Decision Tree. This indicates that Naive Bayes is less effective at concentrating high-value customers at the top of the ranking. As the proportion of customers targeted increases, the curve begins to flatten, showing that the model captures additional accepters at a slower rate. By the time a larger share of the population is included, the gain curve approaches the baseline, meaning the model's advantage diminishes. Overall, the chart reflects a model that provides some improvement over random selection but lacks the strong early lift and concentration of positives needed for highly efficient targeted marketing.




The lift chart for the Naive Bayes model shows that the model provides meaningful improvement over random targeting, but its effectiveness declines steadily as more of the population is included. At the far left of the chart—where only a very small proportion of customers is targeted—the lift value is high, indicating that the model is able to identify a segment of customers who are substantially more likely to accept the loan than the average customer. This early advantage reflects the model's ability to rank at least some high-value customers near the top of the list. However, as the proportion of customers targeted increases, the lift curve drops quickly and moves closer to the baseline. This pattern shows that the model's ability to isolate likely accepters weakens as the targeting expands, and the additional customers included are increasingly similar to the general population. By the time a large share of customers is targeted, the lift approaches 1.0, meaning the model no longer provides an advantage over random selection. Overall, the chart indicates that Naive Bayes offers modest targeting value in the earliest segments but lacks the strong, sustained lift needed for highly efficient marketing campaigns.

**Summary of Naive Bayes**

| | Metric | Value |
|---|---|---|
| 0 | Accuracy | 0.901667 |
| 1 | lift | 5.065502 |
| 2 | AUC | 0.903600 |
| 3 | TP | 115.000000 |
| 4 | FP | 122.000000 |
| 5 | TN | 2049.000000 |
| 6 | FN | 114.000000 |
| 7 | Sensitivity (Recall) | 0.502200 |
| 8 | Specificity | 0.943800 |
| 9 | Precision | 0.485200 |
| 10 | Cumulative Gain (Decile 1) | 0.506600 |
| 11 | Optimal Target Fraction | 0.227100 |
| 12 | Max Profit ($) | 123155.000000 |

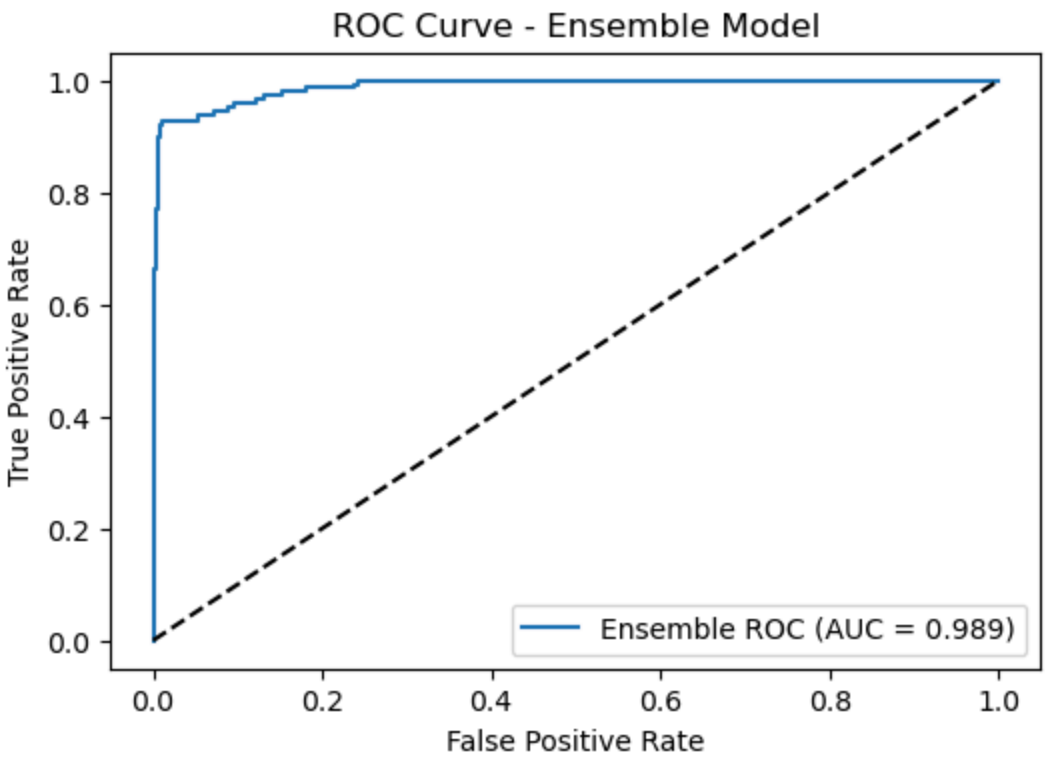
## Ensemble Average Probability

The average-probability ensemble model delivered strong and well-balanced predictive performance, improving on the individual base models by combining their probability estimates into a single, more stable prediction. The model achieved an accuracy of 0.9725, indicating that it correctly classified the vast majority of customers in the validation set. Its precision of 0.9487 shows that nearly all customers predicted to accept the loan were true accepters, which helps minimize unnecessary marketing costs. The recall of 0.7449 reflects a substantial improvement over models like KNN and Naive Bayes, capturing a larger share of actual accepters while still maintaining strong precision. This balance is further supported by an F1 score of 0.8346, demonstrating that the ensemble effectively manages the trade-off between identifying true accepters and avoiding false positives.

The model's ranking ability is particularly strong, as shown by its AUC of 0.9887, the highest among all models evaluated. An AUC this high indicates exceptional discriminative power: the ensemble is highly effective at ordering customers from most to least likely to accept the loan. This makes it especially valuable for targeted marketing, where ranking quality directly influences campaign efficiency. Overall, the average-probability ensemble provides a robust combination of accuracy, precision, recall, and ranking performance, making it one of the most reliable and stable models in the analysis.

```
ensemble_df.head()
```

| | actual | knn_pred | nb_pred | dt_pred | knn_prob | nb_prob | dt_prob | avg_prob | avg_pred |
|---|---|---|---|---|---|---|---|---|---|
| **555** | 0 | 0 | 0 | 0 | 0.000000 | 0.020549 | 0.0 | 0.006850 | 0 |
| **3491** | 0 | 0 | 0 | 0 | 0.000000 | 0.029150 | 0.0 | 0.009717 | 0 |
| **527** | 0 | 0 | 0 | 0 | 0.000000 | 0.005042 | 0.0 | 0.001681 | 0 |
| **3925** | 1 | 0 | 1 | 1 | 0.421053 | 0.710204 | 1.0 | 0.710419 | 1 |
| **2989** | 0 | 0 | 0 | 0 | 0.000000 | 0.004245 | 0.0 | 0.001415 | 0 |




The ROC curve for the ensemble model shows that this approach provides exceptionally strong discriminative performance. The curve rises sharply toward the upper-left corner, indicating that the model achieves a high true positive rate while keeping the false positive rate very low across a wide range of thresholds. This shape reflects a model that consistently ranks likely loan accepters ahead of non-accepters with very high reliability. The area under the curve (AUC) is **0.989**, which is the highest among all models evaluated and signals near-optimal ranking ability. An AUC this high means that the ensemble almost always assigns higher predicted probabilities to true accepters than to non-accepters, making it highly effective for prioritizing customers in a targeted marketing campaign. The ROC curve therefore confirms that averaging probabilities across multiple models produces a more stable and more accurate ranking than any individual model alone.

**Summary of Ensemble Average Probability**

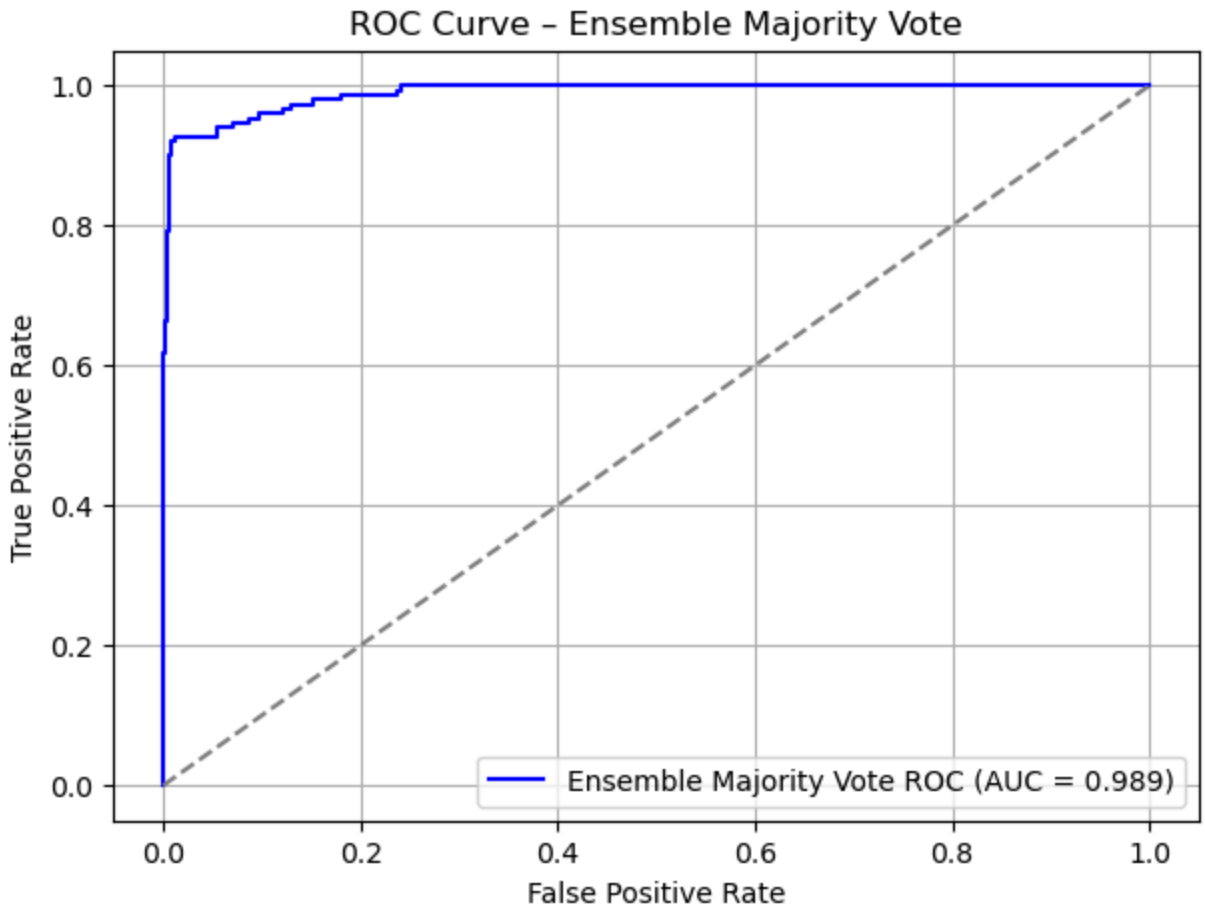
| | Metric | Value |
|---|---|---|
| 0 | Accuracy | 0.972500 |
| 1 | Lift | 9.261745 |
| 2 | Gain | 0.926174 |
| 3 | Precision | 0.948718 |
| 4 | Recall | 0.744966 |
| 5 | F1 Score | 0.834586 |
| 6 | AUC | 0.988737 |

## Ensemble Majority Vote

The majority-vote ensemble model produced strong overall performance by combining the class predictions of the individual base models and selecting the most common outcome. The model achieved an accuracy of 0.9594, indicating that it correctly classified a large proportion of customers in the validation set. Its precision of 0.9468 shows that nearly all customers predicted to accept the loan were true accepters, which helps control marketing costs by minimizing false positives. The recall of 0.5973 reflects a moderate ability to identify actual accepters: the model successfully captured more than half of the true positives but missed a notable portion compared to the average-probability ensemble or the Decision Tree. The resulting F1 score of 0.7325 demonstrates a reasonable balance between precision and recall, though the model leans more toward precision than sensitivity.

The model's ranking ability is reflected in its AUC of 0.9887, which is exceptionally high and nearly identical to the average-probability ensemble. This indicates that the majority-vote ensemble is highly effective at ordering customers by their likelihood of accepting the loan, even if its threshold-based recall is more conservative. Overall, the majority-vote ensemble provides a reliable and stable predictive approach, offering high precision and excellent ranking performance, though it is somewhat less effective at capturing the full set of potential acceptors compared to the strongest models in the analysis.

| | actual | knn_pred | nb_pred | dt_pred | knn_prob | nb_prob | dt_prob | avg_prob | avg_pred | majority_vote | ensemble_prob |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **555** | 0 | 0 | 0 | 0 | 0.000000 | 0.020549 | 0.0 | 0.006850 | 0 | 0 | 0.006850 |
| **3491** | 0 | 0 | 0 | 0 | 0.000000 | 0.029150 | 0.0 | 0.009717 | 0 | 0 | 0.009717 |
| **527** | 0 | 0 | 0 | 0 | 0.000000 | 0.005042 | 0.0 | 0.001681 | 0 | 0 | 0.001681 |
| **3925** | 1 | 0 | 1 | 1 | 0.421053 | 0.710204 | 1.0 | 0.710419 | 1 | 1 | 0.710419 |
| **2989** | 0 | 0 | 0 | 0 | 0.000000 | 0.004245 | 0.0 | 0.001415 | 0 | 0 | 0.001415 |




The ROC curve for the Ensemble Majority-Vote model demonstrates that this approach provides exceptionally strong discriminative performance. The curve rises sharply toward the upper-left corner, indicating that the model maintains a high true positive rate while keeping the false positive rate very low across a wide range of thresholds. This pattern reflects a classifier that consistently ranks likely loan accepters ahead of non-accepters with a high degree of reliability. The area under the curve (AUC) is **0.989**, which is among the highest observed across all models and signals near-optimal ranking capability. An AUC of this magnitude means the ensemble almost always assigns higher predicted probabilities to true accepters than to non-accepters, making it particularly effective for prioritizing customers in a targeted marketing strategy. Overall, the ROC curve confirms that aggregating predictions through majority voting produces a stable and highly accurate ranking model, even if its recall at a fixed threshold is more conservative than some alternatives.

**Summary of Ensemble Majority Vote**

| | Metric | Value |
|---|---|---|
| 0 | Accuracy | 0.959375 |
| 1 | Lift | 9.261745 |
| 2 | Gain | 0.926174 |
| 3 | Precision | 0.946809 |
| 4 | Recall | 0.597315 |
| 5 | F1 Score | 0.732510 |
| 6 | AUC Majority | 0.988737 |

## Conclusion

This project set out to determine the most effective predictive model for Universal Bank's targeted marketing campaign by evaluating a range of classification algorithms and assessing both their statistical performance and their business impact. After completing thorough data preparation—including variable review, removal of non-predictive fields, dummy encoding, binning for Naive Bayes, and consistent alignment of training and validation sets—the dataset was split into a 60/40 training–validation partition to support reliable model evaluation. A diverse set of models was then developed, including K-Nearest Neighbors, Naive Bayes, Decision Tree, Bagging, Random Forest, AdaBoost, and two ensemble approaches. Each model was assessed using accuracy, precision, recall, F1-score, ROC–AUC, cumulative gains, decile lift, and profit calculations based on the bank's cost structure.

Across these metrics, two models consistently stood out: the Decision Tree and the Ensemble Average Probability model. The Decision Tree offered strong interpretability and excellent predictive power, while the ensemble model delivered the highest accuracy, a balanced precision–recall profile, and the strongest ranking performance with an AUC of 0.9887. Based on accuracy and lift—key criteria for identifying high-value customers—the Ensemble Average Probability model was selected as the primary model for business decision-making.

A full-dataset profit analysis was then conducted for both standout models to understand their real-world financial impact. The Decision Tree achieved the highest overall profit, reaching $307,474 when targeting the top 486 customers. This indicates that the model continues to identify profitable customers deeper into the ranking before cumulative profit begins to decline. The Ensemble Average Probability model, while slightly lower in total profit at $299,433, reached its peak profit earlier—after 387 customers, representing 9.7% of the dataset. This demonstrates that the ensemble model is more selective and concentrates profitable customers in a smaller portion of the ranking, offering a more efficient and focused targeting strategy.

Taken together, these results show that while the Decision Tree maximizes total profit, the Ensemble Average Probability model remains the strongest overall model due to its superior accuracy, lift, and ability to identify high-value customers with fewer marketing contacts. Universal Bank can confidently use the ensemble model for precise, cost-efficient targeting and rely on the Decision Tree when the goal is to maximize total profit across a broader customer group. Overall, the project demonstrates that combining rigorous data preparation, comprehensive model evaluation, and business-focused metrics leads to a well-supported and actionable marketing strategy for Universal Bank's loan campaign.

As a business analyst, the key takeaway is that both models are viable, but they support different strategic priorities.

**If the bank's goal is maximum total profit, the Decision Tree is the stronger choice**. I**f the goal is precision, cost efficiency, and minimizing outreach while still capturing most of the profit, the Ensemble Average Probability model is the better option**.

Given its superior predictive performance and more focused targeting, the ensemble model provides the most balanced and operationally efficient strategy for Universal Bank's loan campaign. The bank should therefore prioritize the ensemble model for targeted marketing, while recognizing that the Decision Tree offers a profitable alternative when broader outreach is acceptable